

Day 17: Endpoint Security (EDR & XDR Concepts)

Objective

The objective of this session is to develop a comprehensive understanding of how endpoint security solutions detect, analyze, and respond to cyber threats targeting individual systems such as laptops, desktops, and servers.

By the end of this session, learners will be able to:

- Understand the concept of endpoint security
- Differentiate between EDR and XDR
- Identify suspicious endpoint activities
- Analyze attack scenarios involving endpoints
- Apply detection logic for threat identification
- Understand SOC workflows in endpoint detection and response
- Recommend appropriate response and mitigation strategies

1. Endpoint Security

Endpoint security refers to the practice of protecting endpoint devices such as:

- Laptops
- Desktops
- Servers
- Mobile devices

from cyber threats and unauthorized access.

Key Functions

- Continuous monitoring of endpoint activities
- Detection of malicious processes and files
- Prevention of unauthorized access
- Incident response and containment

Importance of Endpoint Security

- Endpoints are primary targets for attackers
- Protects against malware, ransomware, and exploits
- Prevents lateral movement within a network
- Ensures data security and system integrity

2. Endpoint Detection and Response (EDR)

EDR is a security solution that focuses on monitoring and analyzing endpoint activities to detect and respond to threats in real time.

Core Capabilities

- 1. Monitoring processes**
 - a. Tracks system processes
 - b. Monitors file activity
 - c. Observes user actions
- 2. Detection threats**
 - a. Identifies suspicious behavior
 - b. Detects malware and exploits
 - c. Uses behavioral analysis
- 3. Responds (kill process, isolate system)**
 - a. Kill malicious processes
 - b. Isolate infected systems
 - c. Remove or quarantine files

Key Features

- Real-time visibility into endpoint activity
- Behavioral threat detection
- Incident investigation capabilities
- Automated and manual response actions

3. Extended Detection and Response (XDR)

XDR is an advanced security solution that extends beyond endpoints by integrating multiple data sources for better threat detection.

Data Sources in XDR

- Endpoint logs (EDR)
- Network traffic
- Email systems
- Cloud environments

Key Advantages

- Provides centralized visibility
- Correlates data across multiple layers
- Detects complex, multi-stage attacks

- Reduces false positives

EDR vs XDR

Feature	EDR	XDR
Scope	Endpoints only	Multiple domains (endpoint, network, email)
Visibility	Limited	Broad and centralized
Detection	Endpoint-based	Correlation-based
Complexity	Moderate	Advanced
Use Case	Endpoint threats	Multi-stage attacks

4. Example Scenario Analysis

Alert: malware.exe running

Connected to external IP

Action: Kill process, isolate system

Observed Activity

- Alert: malware.exe is running
- External connection detected to unknown IP
- Suspicious behavior observed

Step-by-Step Analysis

Step 1: Unknown Process Execution

- Process malware.exe is not recognized
- Indicates possible malware infection

Step 2: External Network Connection

- System connects to an unknown external IP
- Possible Command & Control (C2) communication

Step 3: Suspicious Behavior

- Continuous communication with external server
- Potential data exfiltration or remote control

Security Interpretation

- High likelihood of malware infection
- Possible attacker control over the system
- Risk of data theft or system compromise

Risk Level

HIGH

5. Detection Logic

Detection logic helps SOC analysts automatically identify suspicious activities.

Basic Detection Rule

IF unknown process is running
AND system connects to external IP
THEN trigger alert: Possible malware activity

Advanced Detection Logic

IF process_name NOT IN trusted_list
AND external_connection = TRUE
AND IP reputation = suspicious
THEN trigger HIGH severity alert: Endpoint Compromise

Purpose of Detection Logic

- Automates threat detection
- Reduces manual effort
- Improves accuracy
- Enables faster incident response

6. SOC Workflow for Endpoint Security

L1 Analyst (Detection)

- Monitor alerts from EDR/XDR tools
- Identify suspicious endpoint activities
- Escalate potential threats

L2 Analyst (Investigation & Response)

- Analyze process behavior
- Investigate logs and connections
- Confirm malicious activity
- Take response actions (kill process, isolate system)

L3 Analyst (Threat Engineering)

- Improve detection rules
- Develop advanced correlation logic
- Enhance endpoint security strategies

7. Attack Scenario (Assignment)

Scenario

An attacker delivers malware through a phishing email attachment.

Attack Steps

1. User opens malicious attachment
2. Malware (malware.exe) executes on the system
3. Malware establishes connection to external server
4. Attacker gains remote access
5. Sensitive data is accessed or exfiltrated

Attack Classification

- Initial Access: Phishing
- Execution: Malware execution
- Command & Control: External communication
- Impact: Data theft / system compromise

8. Response & Mitigation Strategies

Immediate Actions

- Kill malicious process (malware.exe)
- Isolate affected endpoint from network
- Block suspicious external IP

Containment

- Prevent spread to other systems
- Monitor similar endpoints for infection

Investigation

- Analyze system logs
- Identify source of infection
- Check for persistence mechanisms

Prevention

- Deploy updated EDR/XDR solutions
- Enable real-time monitoring
- Conduct user awareness training
- Restrict execution of unknown files

Final Assessment

- **Attack Type:** Malware Infection
- **Technique:** Execution + Command & Control
- **Impact:**
 - Unauthorized system access
 - Data exfiltration
 - System compromise

1. Explain EDR vs XDR

EDR (Endpoint Detection and Response)

- EDR focuses on monitoring and securing individual endpoints such as laptops, desktops, and servers. It continuously collects endpoint data and detects suspicious activities.

Key Features of EDR:

- Monitors system processes and activities
- Detects threats like malware, ransomware, and unauthorized access
- Provides real-time alerts
- Responds by isolating the system or killing malicious processes
- Focuses only on endpoint-level visibility

XDR (Extended Detection and Response):

- XDR extends beyond endpoints by integrating data from multiple security layers such as endpoints, networks, email, and cloud systems.

Key Features of XDR:

- Correlates data from multiple sources (endpoint + network + email)
- Provides a unified view of threats
- Detects complex, multi-stage attacks
- Reduces alert fatigue through better correlation
- Enables faster and more accurate incident response

Difference Between EDR and XDR:

Feature	EDR	XDR
Scope	Endpoint only	Multiple security layers
Data Sources	Endpoint logs	Endpoint + Network + Email + Cloud
Detection	Single-point detection	Correlated multi-source detection
Visibility	Limited	Broad and unified
Use Case	Endpoint protection	Advanced threat detection & response

2. Write Attack Scenario

Scenario: Malware Infection with External Communication

Step 1: Initial Infection

- A user downloads a malicious file named **malware.exe** from an untrusted source (email attachment or website).

Step 2: Execution

- The file is executed on the system.
- A suspicious process (**malware.exe**) starts running in the background.

Step 3: Persistence

- The malware attempts to maintain access by creating registry entries or scheduled tasks.

Step 4: External Communication

- The infected system connects to an unknown external IP address.
- This may be a Command & Control (C2) server.

Step 5: Malicious Activity

- The attacker may:
 - Steal sensitive data
 - Download additional payloads
 - Execute remote commands

Detection Insight:

- Unknown process + external connection = strong indicator of compromise

3. Suggest Response

Immediate Response Actions:

- Kill the malicious process (**malware.exe**)
- Isolate the infected endpoint from the network
- Block the malicious external IP address
- Disable affected user account (if needed)

Investigation Steps:

- Analyze endpoint logs to trace execution timeline
- Identify how the malware entered the system
- Check for persistence mechanisms
- Review network logs for suspicious connections
- Determine if lateral movement occurred

Containment:

- Quarantine affected systems
- Monitor other endpoints for similar indicators
- Prevent further spread

Eradication:

- Remove malware completely from the system
- Delete malicious files and registry entries
- Apply security patches

Recovery:

- Restore system from a clean backup
- Re-enable system access after validation

Prevention Measures:

- Enable Endpoint Detection and Response (EDR)
- Implement Extended Detection and Response (XDR)
- Use Multi-Factor Authentication (MFA)
- Conduct user awareness training
- Block suspicious file downloads
- Improve detection rules in SIEM

Final Summary

- **EDR** protects endpoints by detecting and responding to threats locally.
- **XDR** provides broader visibility by correlating data across multiple systems.
- Combining both improves detection of advanced, multi-stage cyber attacks.

Key Takeaway

Endpoint security is critical in modern cybersecurity. EDR provides deep visibility at the endpoint level, while XDR enhances detection by correlating data across multiple sources. Together, they enable faster detection, accurate analysis, and effective response to cyber threats.